AI ACADEMY · LEVEL 4 · GRADE 9 · WEEK 29 · STUDENT LESSON

Feature Engineering

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 12: units affect feature calculations.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Create features available at prediction time. Try the recall before opening notes.

Your learning target

I can create features available at **prediction** time.

I can fit feature **transformation**s using **training evidence** only.

I can compare a feature change without changing the evaluation set.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Feature engineering transforms raw **input**s into useful structure while preserving **prediction**-time **availability** and avoiding **leakage**.



Your target

LEARNING TARGET Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

Builder name and experiment date:

Readiness for the connected explanation

Why must pre**process**ing be fitted on **training** only?

Readiness for Relationships and confounding

Can a survey of only one club establish the preference of all students?

My readiness explanation

This week's end goal

By the end, I can explain **feature engineering** using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Feature Engineering case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **feature engineering** to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

A useful feature represents information available when the **prediction** is made. In a fictional parcel task, length and width are measured before packing, so area=length*width is available.

The final delivery delay is not available when predicting dispatch time. A strong correlation cannot make a future value legitimate.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **feature engineering**.

Apply the connected idea: Relationships and confounding

In a fictional log, groups that practiced longer also scored higher: 1 hour with 4 points, 2 hours with 6 and 3 hours with 8. This pattern is an **association**. Prior experience could affect both practice choices and scores.

The table alone cannot establish the effect of adding practice time.

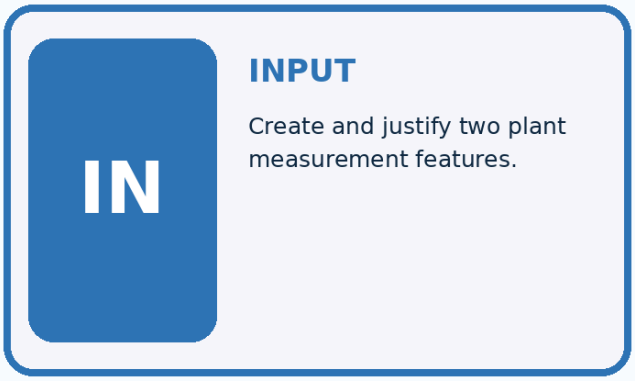
See how it works

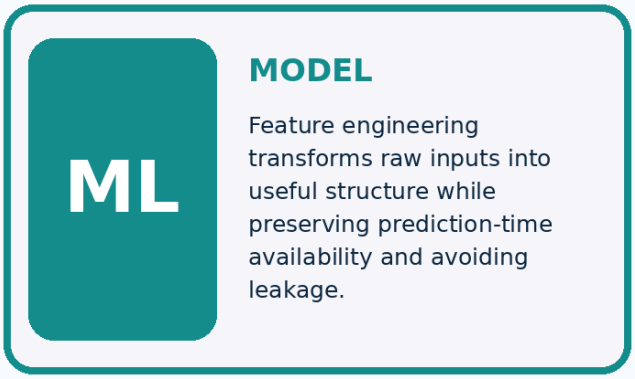
> Follow the mechanism for Feature Engineering. Connect each labelled input, deciding step and result.

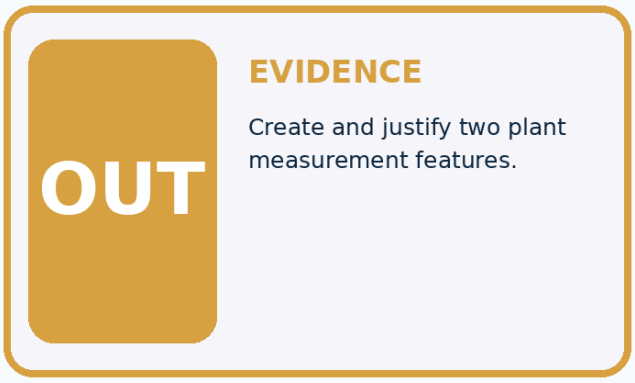
ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Create and justify two plant measurement features.

MODEL IDEA **Feature engineering** transforms raw **input**s into useful structure while preserving **prediction**-time **availability** and avoiding **leakage**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Raw leaf length=8 cm and width=4 cm. Derived aspect_ratio = 4 ÷ 8 = 0.5.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Training rectangles have lengths [2, 4, 6] and widths [3, 2, 1], so areas are[6, 8, 6] square cm. **Training** mean area is 20/3. A validation rectangle length 5 width 2 has area 10 and centered area 10-20/3=10/3.

Freeze the **training** mean; do not recompute it with validation.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Apply the connected idea: Relationships and confounding

```
rows = [{"hours": 1, "score": 4},
        {"hours": 2, "score": 6},
        {"hours": 3, "score": 8}]
for row in rows:
    print(row["hours"], row["score"])
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Observed	More practice with higher scores	Association
Possible other factor	Prior experience	May affect both
Missing evidence	Comparable starting conditions	Limits causal claim

Read each row from left to right and connect the evidence to the stated rule.

The numbers rise together in these three invented records. A description such as higher practice hours went with higher scores is supported. A claim that one extra hour guarantees two more points goes beyond the **observation**s.

No statistical calculation is required to see this limit.

Separate a pattern from a cause

The three records show that higher practice time goes with higher scores. They do not tell us what would happen if the same learner practised for one more hour.

Prior experience could make someone both practise more and score higher. That is a possible **confounder**: another influence connected to both quantities. First describe the pattern accurately.

Then propose a **comparison** that addresses alternative explanations rather than treating the pattern as a guarantee.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

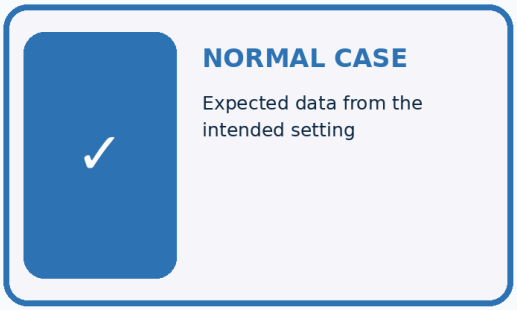
Compare the cases

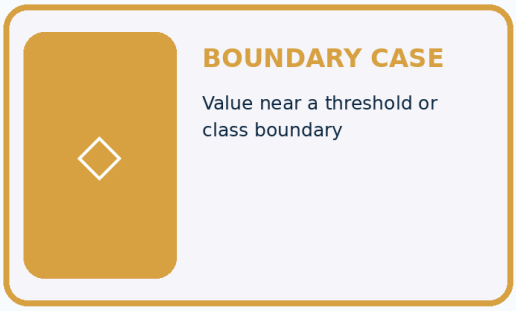
> Compare the supplied cases for Feature Engineering. Identify the detail that changes the conclusion.

EVALUATION DESIGN

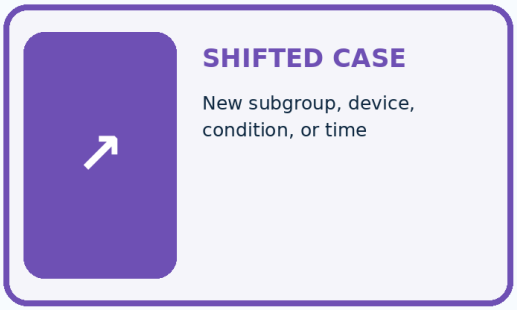
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Include the query in the centering mean.

Predicted result

Prediction: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Why the result follows

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

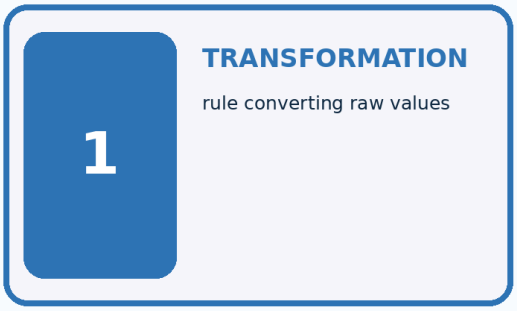
> Use the vocabulary of Feature Engineering to explain a concrete decision, rather than only

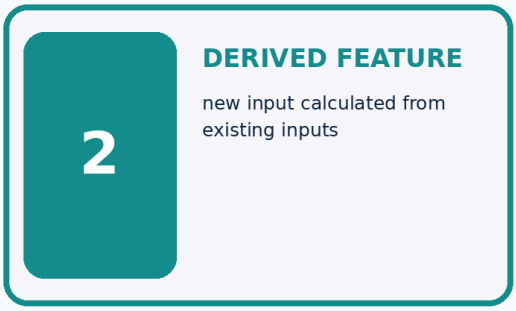
> memorising definitions.

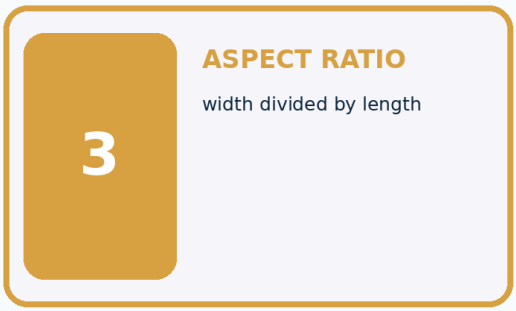
VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.









Term	Precise meaning	My ML example
transformation	rule converting raw values	
derived feature	new input calculated from existing inputs	
aspect ratio	width divided by length	
feature pipeline	fixed ordered transformations	

MISCONCEPTION TO REPAIR Creating many features automatically gives the model more useful information.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Feature engineering	Constructing model inputs from available measurements
Derived feature	An input computed from other inputs
Leakage	Information entering a model that would not be available legitimately
Availability	Whether a value exists when a prediction is needed

Terms for Relationships and confounding

Term	Meaning
Association	Two variables vary together in observed data
Cause	A change in one factor produces a change in another
Confounder	A factor related to both the proposed cause and outcome
Comparison	Examining outcomes under different conditions

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Feature Engineering, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 29 • WORKED EVIDENCE

1. Raw leaf length=8 cm and width=4 cm.
2. Derived aspect_ratio = 4 ÷ 8 = 0.5.
3. Color histogram summarizes pixel ranges without using the target label.
4. Fit transformation settings on training data only; document unit, purpose, availability, and leakage check.

Step 1

STEP 1 Raw leaf length=8 cm and width=4 cm.

Step 2

STEP 2 Derived aspect_ratio = 4 ÷ 8 = 0.5.

Step 3

STEP 3 Color histogram summarizes pixel ranges without using the target label.

Step 4

STEP 4 Fit **transformation** settings on **training** data only; document unit, purpose, **availability**, and **leakage** check.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Feature engineering at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Raw measurements	Length and width	Available before prediction
Derived area	Multiply length by width	Units square cm
Centering	Subtract training mean	Reuse frozen value

Area is a deterministic transformation; it adds a representation, not a new measured fact. It may help a limited model express a relationship, or may do nothing useful. Compare original versus extended features under the same split, metric and development policy.

Center the validation area using the frozen training mean. Explain the steps or evidence behind your response.

Apply the connected idea: Relationships and confounding

Name one factor besides practice duration that could influence both time chosen and score.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Feature Engineering

Engineer a feature with a **transformation** defined before evaluating a new leaf.

The facts we will use

Supplied facts

Training rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/29

Feature Engineering

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Construct area

Query area=2 x 4=8 cm2.



2. Fit centering

Training mean=(2+6+10)/3=6 cm2.



3. Apply unchanged transform

Centered query feature=8-6=2 cm2.

CHANGE ONE THING

Include the query in the centering mean.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Construct area	Query area=2×4=8 cm².
2. Fit centering	Training mean=(2+6+10)/3=6 cm².
3. Apply unchanged transform	Centered query feature=8-6=2 cm².

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Area is a geometric toy feature. Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

Keep the stated limit

Read the labelled picture: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Include the query in the centering mean. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Feature Engineering claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: Creating many features automatically gives the model more useful information. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE Creating many features automatically gives the model more useful information.

Diagnose one **cause** and name the cleanest next experiment:

A closer explanation

An identifier may appear predictive be**cause** related objects leak across splits. Inspect grouped records and proxy risks before adding it.

Never include private student details for this exercise; use the supplied fictional measurements.

How would you test adding area without claiming it must help?

Apply the connected idea: Relationships and confounding

A fairer **comparison** could offer the same task, instructions and assessment and randomly assign fictional groups to practice conditions.

In real classrooms, a teacher must approve any activity and protect equal learning opportunities; this lesson uses a paper plan.

Even a better comparison has only four fictional groups. What should be reported?

STUDENT LESSON · 8 OF 18

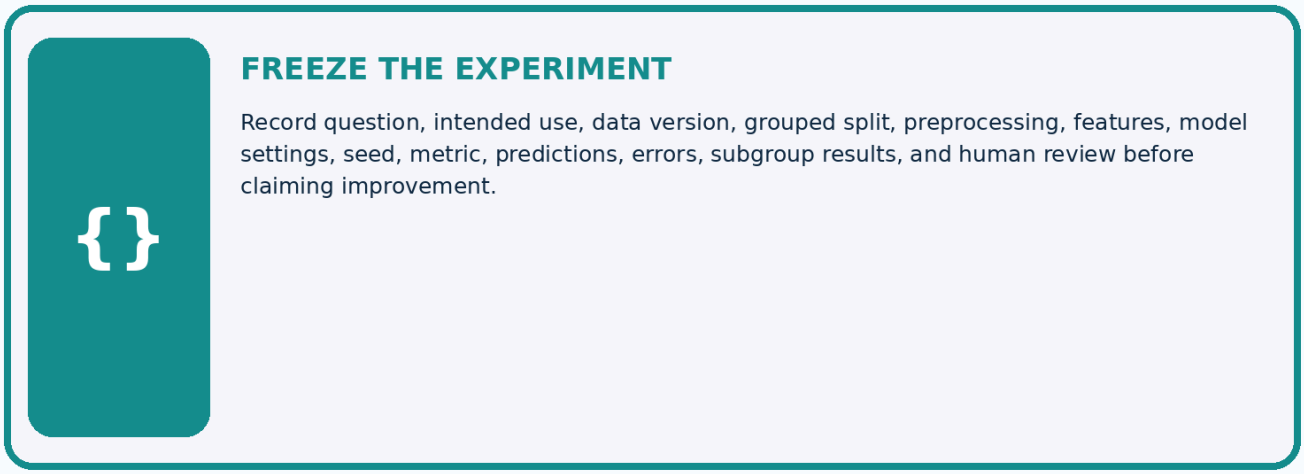
Practise together

> Practise Feature Engineering with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

Question to explain

QUESTION + intended use + success threshold

DATA version + grouped split + **leakage** check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Build a feature log with name, units, source, time available, **transformation** and missing-value policy. If width is missing, follow a declared **training**-derived imputation or return a clear unavailable result.

Do not silently use zero, which describes a zero-area object.

Apply the connected idea: Relationships and confounding

A chart title Practice **cause**s success overstates the **evidence**. Replace it with Practice time and scores in three fictional records, and describe the possible **confounder**.

Better wording does not repair the study, but it makes the report honest.

Practise with less help: Transform validation

Use workbook W2 for the connected example "**Feature engineering**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Feature engineering**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Relationships and confounding

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

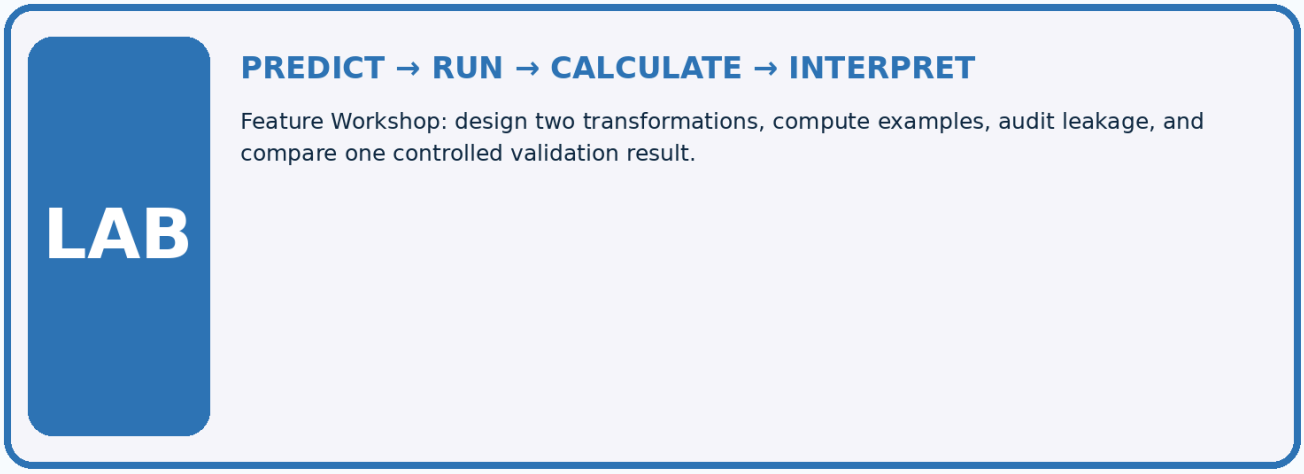
Run the investigation

> Investigate Feature Engineering. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.



1. Write the hypothesis and expected **evidence** before running.

2. Feature Workshop: design two **transformation**s, compute examples, audit **leakage**, and compare one controlled validation result.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____
> **Meaning / next test:** _____

> **Case:** Boundary
> **Prediction:** _____
> **Actual:** _____
> **Calculation / evidence:** _____

> **Meaning / next test:** _____

> **Case:** Missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

> **Case:** Shifted
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Calculation / evidence:** _____
>
> **Meaning / next test:** _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

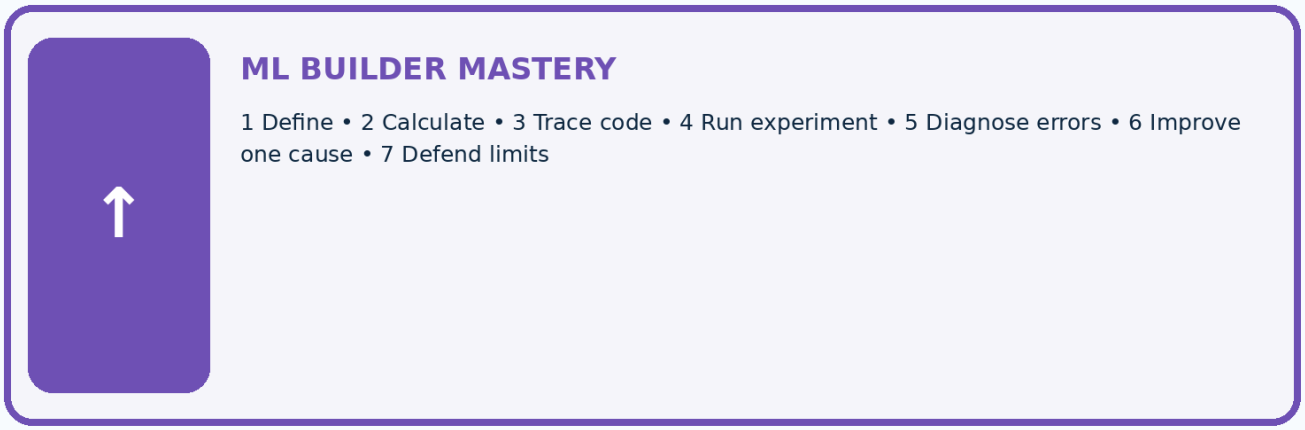
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Feature Engineering. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **transformation**, **derived feature**, **aspect ratio**, **feature pipeline**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Creating many features automatically gives the model more useful information.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one **cause** and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is

completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.



THIS WEEK'S CHECKPOINT Add only justified Plant Classifier features, with formula, source fields, expected benefit, and **leakage** test.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

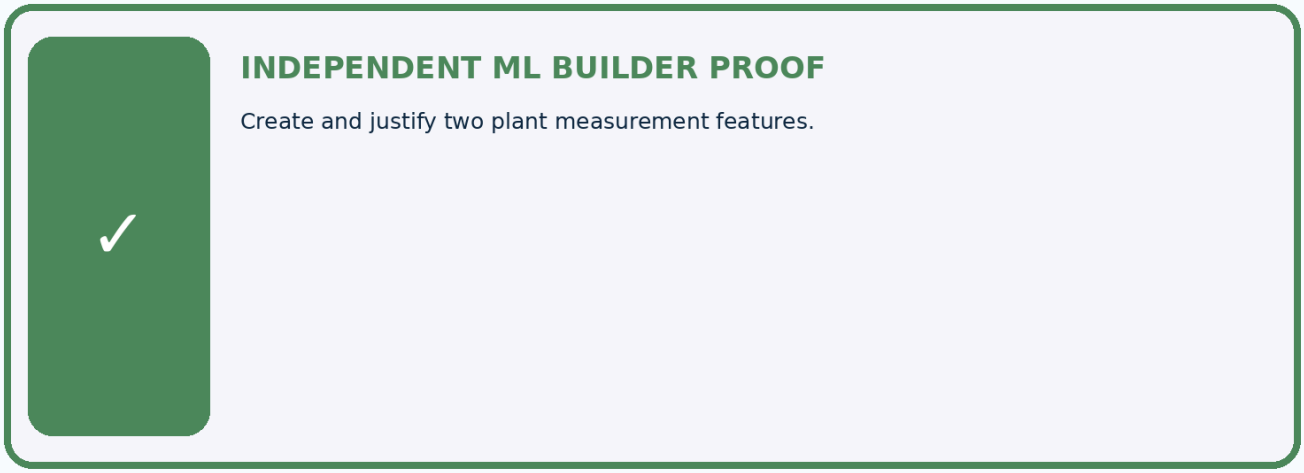
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Create and justify two plant measurement features.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

> **Connect to the next lesson**
>
> NEXT BRIDGE Carry the experiment **evidence** into: Improve Generalization.

Study a complete case

> Revisit the complete Feature Engineering case for supported practice. Repeated examples do not become fresh assessments.

Week 29 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Features are leaf length and width in cm; add length/width.

The question

Compute for8 and2; state boundary checks.

Worked explanation

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Engineer a ratio with boundary checks

Evidence or stage	Value or result
Length	8 cm
Width	2 cm
Length divided by width	4, unitless
Width zero or missing	Do not divide, handle explicitly

Features are leaf length and width in cm; add length/width.

Compute for8 and2; state boundary checks.

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 29 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

> A feature is leaf_length divided by leaf_width. A row has width zero. What should the pre**process**ing rule do before division?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and **leakage** test.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Feature Engineering, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Engineer a feature with a **transformation** defined before evaluating a new leaf.

By the end, I can	How I will show it
Create features available at prediction time	Explain the deciding step, show a trace or test, and state one limit.
Fit feature transformations using training evidence only	Explain the deciding step, show a trace or test, and state one limit.
Compare a feature change without changing the evaluation set	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Feature Engineering without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Create features available at **prediction** time

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Engineer a feature with a **transformation** defined before evaluating a new leaf.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Centered query feature=8-6=2 cm².

1. Construct area

2. Fit centering

3. Apply unchanged transform

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Apply unchanged transform Centered query feature=8-6=2 cm². Read the picture in order. Centered query feature=8-6=2 cm².

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Include the query in the centering mean. Which conclusion is supported by the supplied facts?

1. Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The

answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Repair a missing building block

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What **comparison** is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Hold Out Test Data (ai-4/23). Goal: Freeze the full evaluation procedure before testing. Short practice: What is needed for an honest final claim about the revised version?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

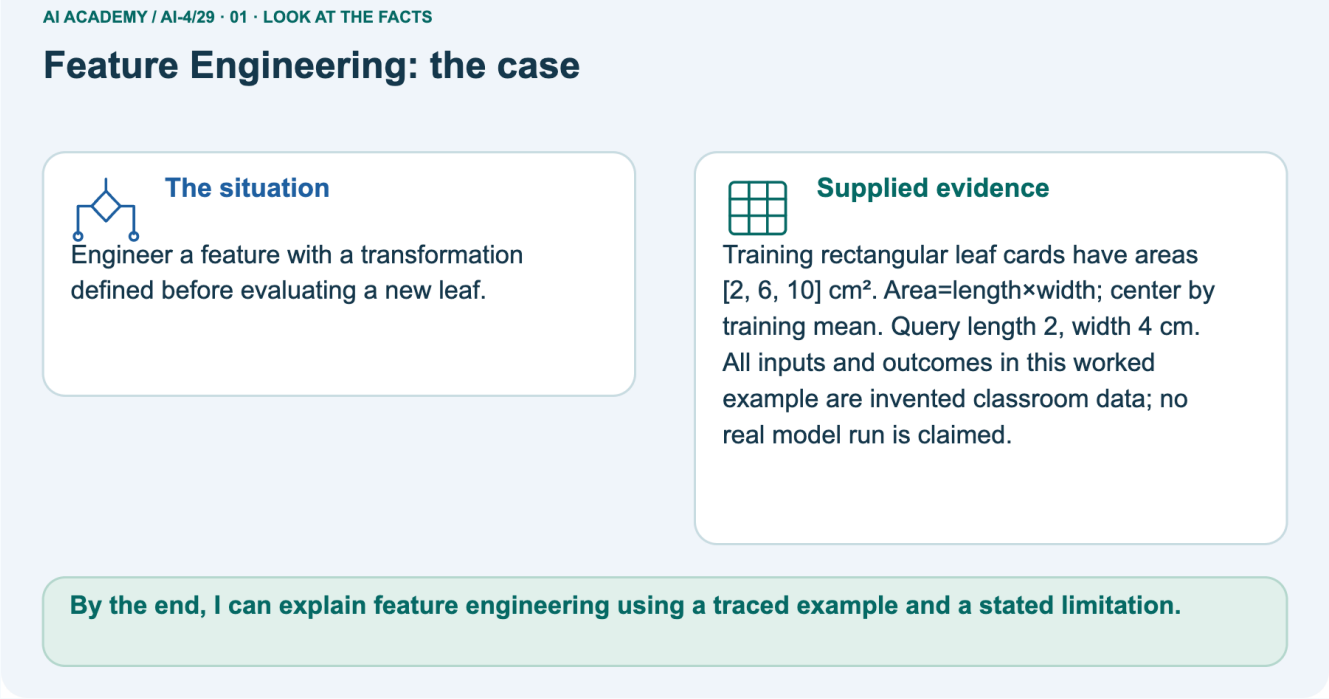
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Feature Engineering graphic using its labels, deciding evidence and relevant limitation.

01. Feature Engineering: the case



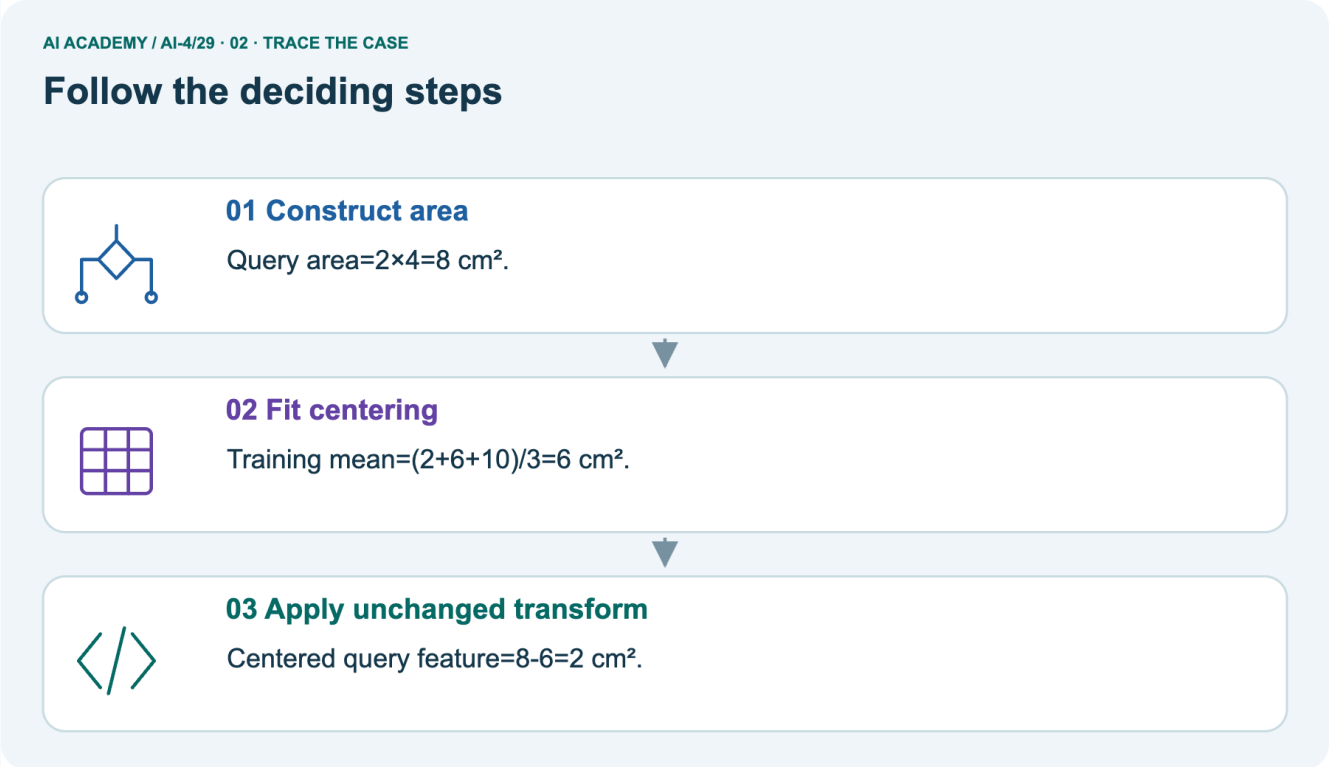
Follow the labelled picture with the matching explanation. Open the vector diagram

Engineer a feature with a **transformation** defined before evaluating a new leaf. Supplied facts: **Training** rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

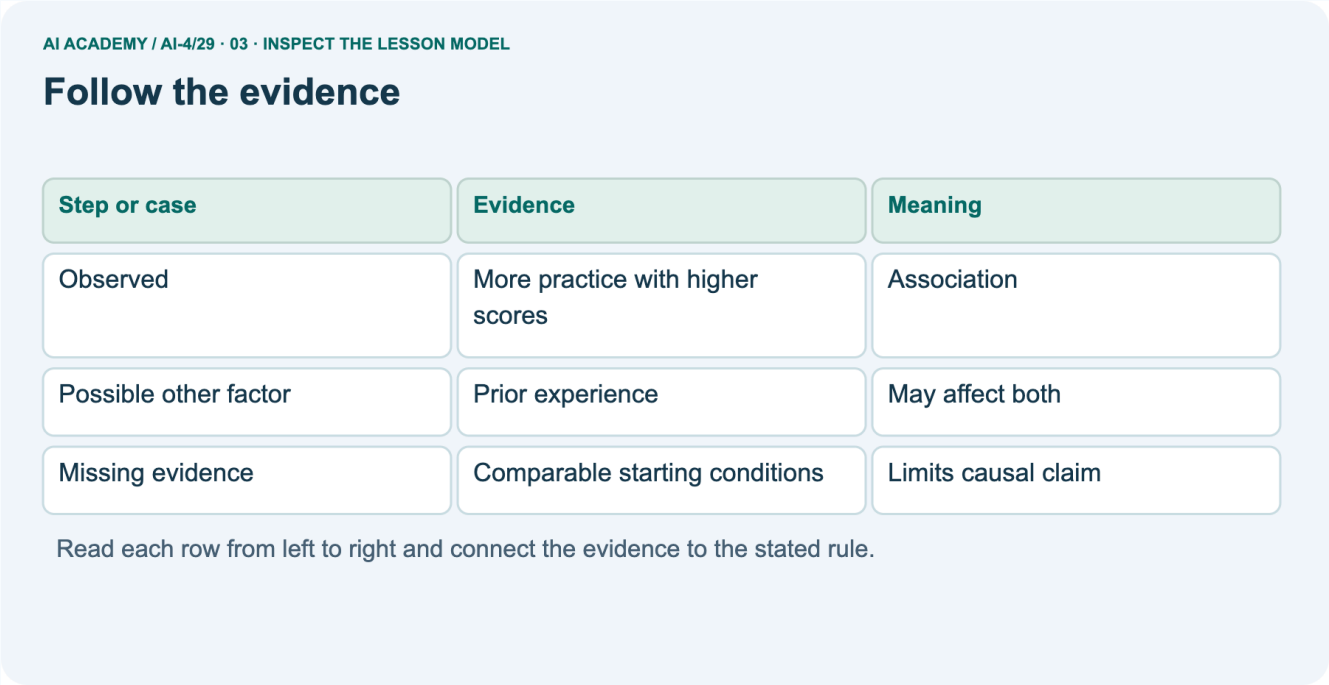


Follow the labelled picture with the matching explanation. Open the vector diagram

Construct area: Query area=2×4=8 cm². → Fit centering: **Training** mean=(2+6+10)/3=6 cm². → Apply unchanged transform: Centered query feature=8-6=2 cm².

> Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

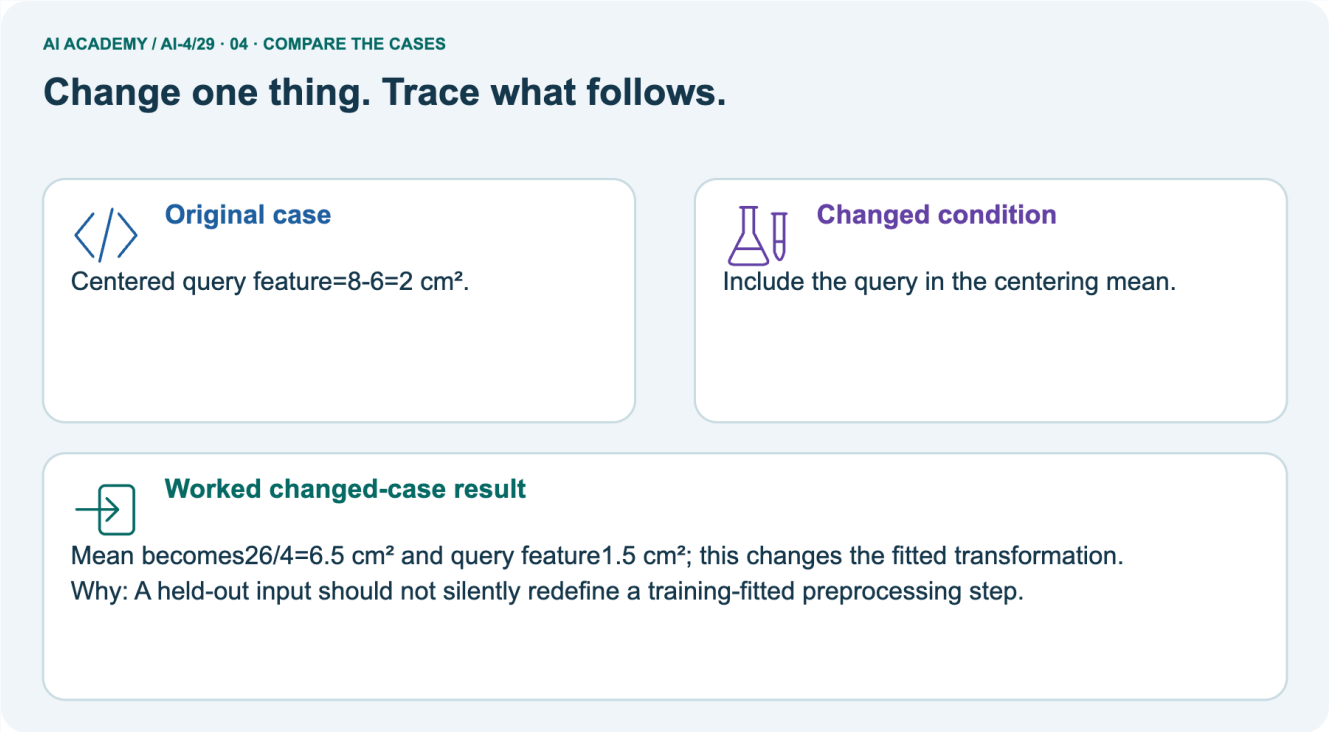


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Observed | More practice with higher scores | **Association**. Possible other factor | Prior experience | May affect both. Missing **evidence** | Comparable starting conditions | Limits causal claim

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



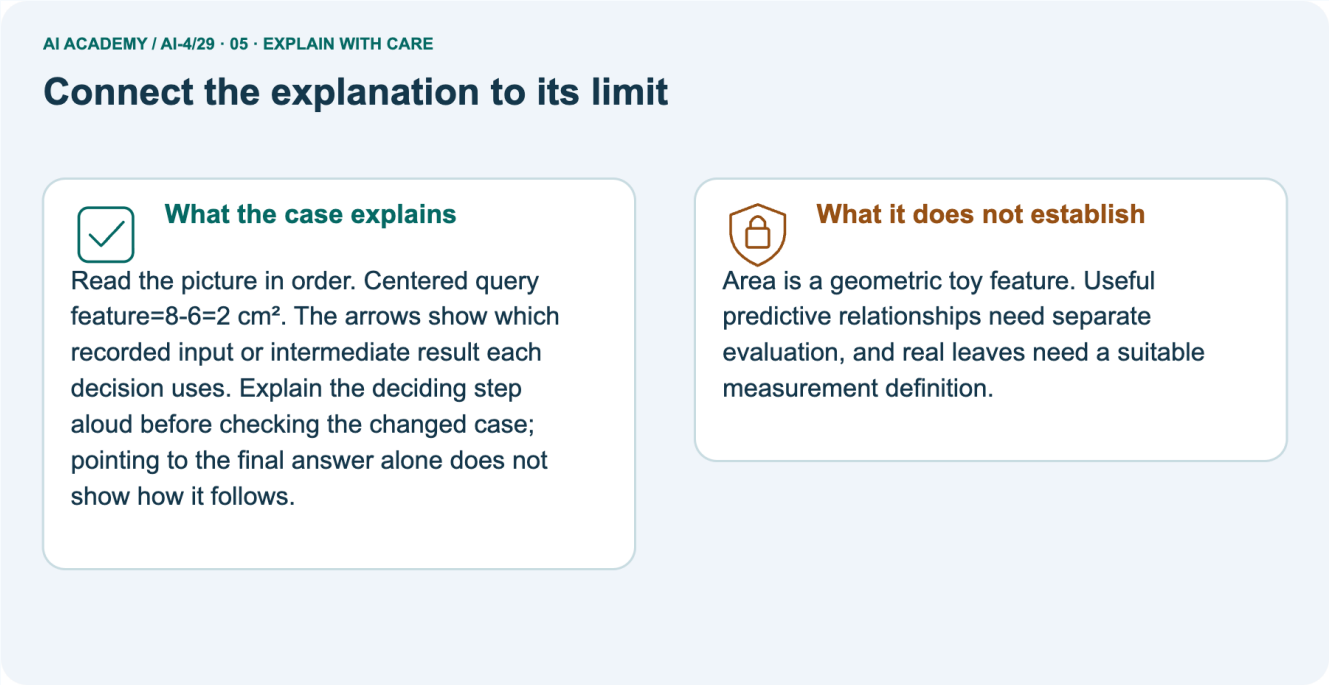
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Centered query feature=8-6=2 cm². Changed condition: Include the query in the centering mean. Expected result: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

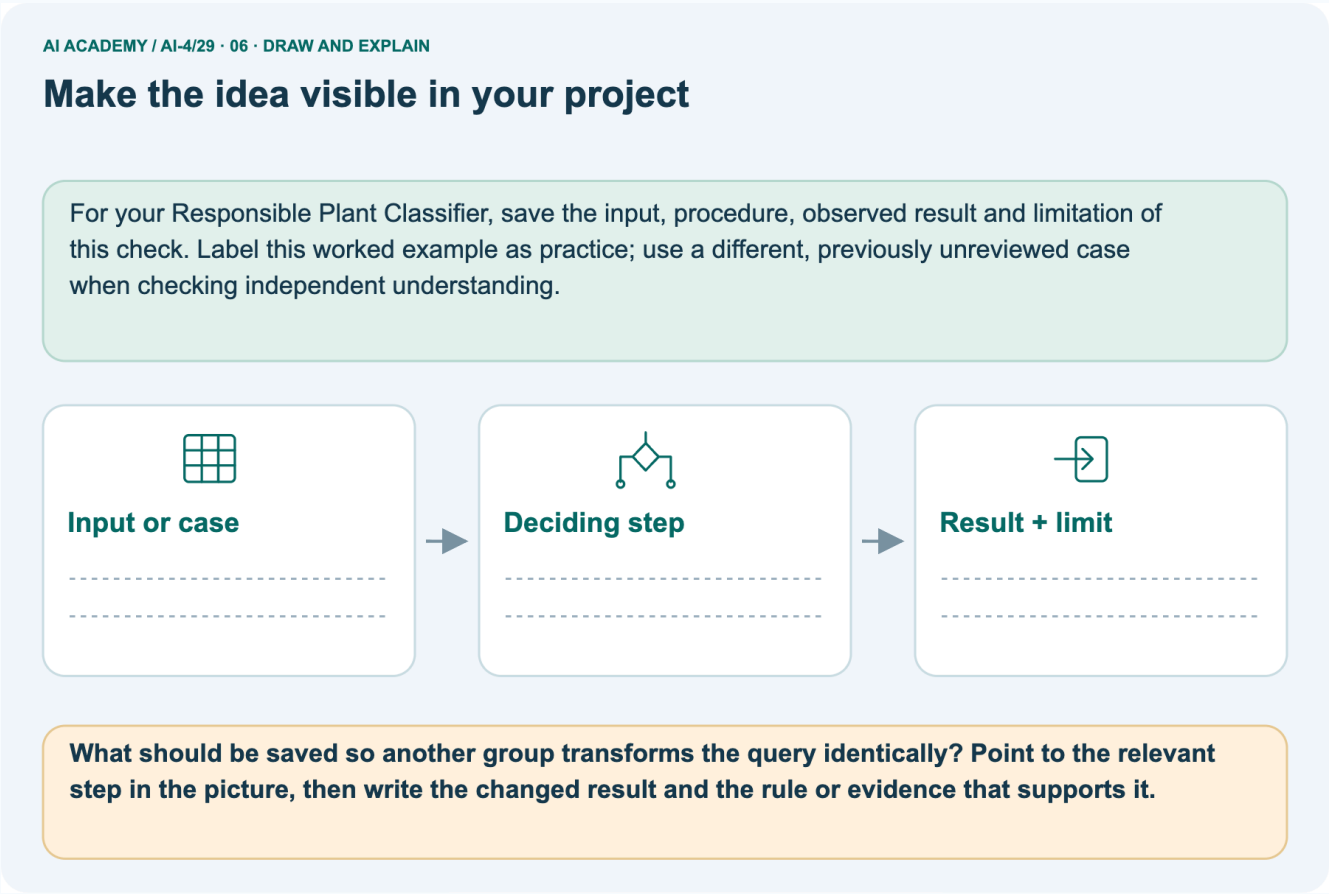
Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Area is a geometric toy feature.

Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.